Python for GenAI

Basics of Python

Note: Prefer Gemini AI Studio, Groq, and TogetherAI APIs for free usage.

1. **Travel Recommendation Bot**:

    - **Inputs**: Start Date, End Date, Preferred Locations (e.g., beach, city, mountains), and Budget Range.
    - **Task**: Create a chatbot that provides personalized travel recommendations based on the provided inputs. Experiment with `temperature` and `top_k` values to see how the recommendations vary in creativity and relevance.

2. **Customer Service Assistant**:

    - **Inputs**: Problem Description, Device Type (e.g., smartphone, laptop), and Urgency Level.
    - **Task**: Develop a bot that assists users with tech troubleshooting. Adjust the `temperature` setting to observe how it influences the tone and assertiveness of the bot's responses, depending on the urgency of the problem.

3. **Job Interview Simulator**:

    - **Inputs**: Job Role (e.g., software engineer, data analyst), Years of Experience, and Key Skills.
    - **Task**: Design a system that simulates a job interview. Experiment with the `top_p` parameter to see how it affects the diversity of interview questions, tailoring them to the provided job role and skills.

4. **Recipe Recommendation Bot**:

    - **Inputs**: List of Available Ingredients, Dietary Restrictions (e.g., vegetarian, gluten-free), and Desired Meal Type (e.g., breakfast, dinner).
    - **Task**: Create a bot that suggests recipes based on the ingredients provided. Modify the `top_k` and `temperature` settings to generate a mix of traditional and innovative recipe ideas.

5. **Fictional Story Generator**:

- **Task**: Develop a bot where users can continue a fictional story by providing prompts. Experiment with `temperature` and `top_p` values to see how they influence the creativity and coherence of the story's continuation.

6. **Q&A Bot for AI and ML**:

    - **Task**: Build a Q&A bot that answers questions related to AI and machine learning. Modify the conversation history length and analyze how it impacts the bot's ability to maintain context and provide relevant answers.

7. **Virtual Python Tutor**:

    - **Task**: Create a virtual tutor that teaches Python programming. Adjust the `top_k` and `temperature` settings to balance between providing clear explanations and offering varied examples that encourage creative problem-solving.

8. **Mental Health Support Chatbot**:

    - **Task**: Develop a chatbot that offers mental health support. Experiment with varying the `temperature` and `top_k` values to generate responses that range from empathetic to more direct, depending on the user's input.

9. **Day Planner Assistant**:

    - **Task**: Create a bot that helps users plan their day by suggesting activities based on the time of day and weather. Adjust the conversation history length to see how it influences the bot's ability to provide consistent and contextually relevant suggestions.

10. **Educational Quiz Bot**:

    - **Task**: Build an educational chatbot that quizzes users on general knowledge topics. Modify the `top_p` and `temperature` values to generate questions of varying difficulty, observing how these changes impact the challenge level of the quiz.